



Crypto Tracker App – Project Report



Project Title:

Real-Time Crypto Price Tracker using React & Redux Toolkit



Project Objective:

To build a **real-time, responsive cryptocurrency price tracker** that mimics platforms like CoinMarketCap, showing live price updates, percent changes, and 24h volume. The project uses **mock data** and simulates **WebSocket behavior** using `setInterval`—all managed through **Redux Toolkit** for clean and scalable state management.



Thought Process & Planning:

1. Component Planning:

- Break down the app into components like:
 - `CryptoTable`: displays all coins
 - `Price`: updates prices with color transitions
 - `Chart`: displays static 7D chart
- Keep each component lightweight and reusable.

2. State Management:

- Use **Redux Toolkit** (`createSlice`) for centralized and reactive state updates.
- Avoid local component state—everything is managed globally.

3. WebSocket Simulation:

- Use `setInterval` in `useEffect` to simulate live updates.
- Randomize `price`, `1h%`, `24h%`, `7d%`, and `volume_24h`.

4. Performance Optimization:

- Use `useSelector` with custom selectors to pull only required data.
- Use `useRef` to track previous prices and avoid unnecessary re-renders.

5. UI/UX Goals:

- Design a responsive table with color-coded fields.
- Provide clean visual cues for increases (green) or decreases (red).

Technology Stack:

- **React** – Component-based architecture
- **Redux Toolkit** – Efficient and structured state management
- **JavaScript (ES6)** – Core logic and updates
- **Tailwind CSS / CSS** – Responsive design (optional)
- **Mock Data** – Static sample assets simulating real coins
- **Hooks** – `useEffect`, `useSelector`, `useDispatch`, `useRef`

Key Functionalities:

- Live update simulation every 3 seconds
- Smart change detection using `useRef`
- Color-coded UI changes (green/red/neutral)
- Lightweight, scalable structure